Q16 NONZERO-TRACE DEGREE-THREE COVERS

Exact curve maps, subgroup transfer, bounded census, and same-target cost | 28 October 2026

Finding. Two Frobenius orbits of elliptic curves over F_(16^3) have exact degree-three covers from cataloged ordinary genus-three curves over F_16, although the elliptic coefficient b has nonzero relative trace. Both representative 409-order subgroup maps are nonzero and both complete relation solves recover 7. Their measured full transfer is more expensive than rho. The exact cover instances are candidates for novelty; a new attack method or measured vulnerability is not established.

Bounded catalog screen

The frozen LMFDB export contains 126 ordinary genus-three Weil-restriction-shaped classes and 10,664 equations. All 24 branch-supported degree-three exponent patterns were tested for each model: 255,936 model-pattern pairs. Every residue system has full column rank 24. Sixty exact identities survived, representing 30 distinct source equations in six classes. Of those descriptions, 24 have trace-zero b, 12 have b=1, and 24 have nonzero-trace b outside the trace-zero locus and b outside the base field. This is a complete screen for the displayed catalog and rational-function ansatz, not all GHS covers.

Exact cover criterion and bound

For C: y^2+h(x)y=f(x), let r_i be the four roots of separable h in K=F_(16^3), and choose an exponent permutation e_i of (2,1,-2,-1). Set U=product (x-r_i)^e_i, X=lambda U, beta=lambda mu, b=beta^2. If f/h^2 + X + beta/X = H^2+H+u in K(x), then Y=X(y/h+H)+beta gives a degree-three map C_K to E: Y^2+XY=X^3+uX^2+b. Four nonzero reduced branch residues force every branch root to be a zero or pole of X. Other degree-three branch patterns have an unremovable order-three pole. With full-rank residue matrix, at most one (lambda,mu) pair can occur for each fixed model and pattern.

Two explicit 409-order subgroup representatives

Here K=F_2[w]/(w^12+w^7+w^6+w^5+w^3+w+1), k=F_2[a]/(a^4+a+1), and the embedding sends a to K mask 1820. Integer masks give polynomial-basis coefficients. Both source curves are E: Y^2+XY=X^3+uX^2+b; both have order 4090=10*409. The exact H numerator and denominator are in the map receipts.

LMFDB model	Base curve C: y^2+h y=f	(u,b), trace(b)	(lambda,mu)
3.16.a_a_ah / 5	y^2+(a^2*x^4+a*x)*y=x^8+(a+1)*x^3+(a^3+a^2+1)*x^2+x+a^3	(1571,1614), 3363	(2925,280)
3.16.a_a_ah / 12	y^2+(a^3*x^4+a*x)*y=x^8+a*x^3+(a^3+a^2+a)*x^2+a^3*x+a^3+1	(3100,2952), 3362	(1706,3295)

Model	G and Q=7G	Image of G (Mumford u;v)	Relations / steps	Full / rho mean	Ratio
5	[3802, 3009]; [952, 2216]	[2622, 1]; [1641]	12 / 100	0.709 s / 0.496 ms	1430x
12	[3047, 101]; [2425, 3200]	[3363, 1641, 3158, 1]; [1641, 1641, 372]	10 / 92	0.828 s / 0.669 ms	1236x

The map pulls back P-2P and adds the three Frobenius conjugates. Both images are nonzero, have order 409, are defined over k, and map Q to seven times the image of G. Each full cost includes construction, transfer, factor-base setup, relation collection, and linear solve. Rho is the mean of 1,000 independent same-target solves in Sage. The relation rows and rho step samples are retained in the receipts.

Prior work and limits

Joux-Vitse (EUROCRYPT 2012, Appendix A.2) publishes characteristic-two trace-zero genus-three covers. General GHS conorm maps and cover/decomposition attacks are earlier work. The displayed b values have nonzero relative trace, so that specific trace-zero construction does not explain these maps. Search did not establish that the exact equations are absent from all publications. The Lean checks cover finite accounting and group-order arithmetic; Sage checks the exact identities, finite-field map, and attack. No global classification of vulnerable curves follows from this census.

Sources: Joux-Vitse, eprint.iacr.org/2011/020.pdf; Magma handbook, math.uzh.ch/sepp/magma-2.25.2-ds/HandbookVolume10.pdf; Iijima et al. (2005), doi:10.1093/ietfec/e88-a.1.97; LMFDB classes 3.16.a_a_ah and 3.16.a_a_h. Companion REPORT.md and JSON receipts retain the exact input, provenance, maps, and measured costs.